

Congruences

Problem 1. Find all positive integers a, b, c such that

$$a \equiv b \pmod{c}, \quad b \equiv c \pmod{a}, \quad c \equiv a \pmod{b}.$$

Problem 2. Let $n \geq 5$ be a squarefree integer. Prove that there exists an integer m such that

$$m^4(m^4 - 5) \equiv 36 \pmod{n}.$$

Problem 3. Let n be a positive integer. Determine, in terms of n , the greatest integer which divides every number of the form $p + 1$, where $p \equiv 2 \pmod{3}$ is a prime number which does not divide n .

Problem 4. Let $N \geq 2$ be a positive integer. Given a sequence of natural numbers $a_1, a_2, \dots, a_{N+1}$ such that for all integers $1 \leq i \leq j \leq N + 1$,

$$a_i a_{i+1} a_{i+2} \cdots a_j \not\equiv 1 \pmod{N}.$$

Prove that there exists a positive integer $k \leq N + 1$ such that $\gcd(a_k, N) \neq 1$.

Problem 5. Find all prime numbers p, q for which the congruence

$$\alpha^{3pq} \equiv \alpha \pmod{3pq}$$

holds for all integers α .

Problem 6. Let p be a prime number. Prove that

$$\sum_{j=0}^p \binom{p}{j} \binom{p+j}{j} \equiv 2^p + 1 \pmod{p^2}.$$

Problem 7. Prove that there exist infinitely many composite numbers n such that

$$3^{n-1} \equiv 2^{n-1} \pmod{n}.$$

Problem 8. Prove that for any positive real number λ , there are n positive numbers $a_1, a_2, \dots, a_n$ ($n \geq 2$), so that $a_1 < a_2 < \cdots < a_n < 2^n \lambda$ and for any $k = 1, 2, \dots, n$ we have

$$\gcd(a_1, a_k) + \gcd(a_2, a_k) + \cdots + \gcd(a_n, a_k) \equiv 0 \pmod{a_k}.$$